RUNI – Algorithms – Homework 5

Due date: Sunday 21/12/2025

NOTE: the general homework guidelines published in homework 1 are valid for all homework assignments in this course.

Problem 1 (24 pts):

Let $G = (V, E)$ be an undirected graph. A set $S \subseteq V$ is said to be independent if for every $u, v \in S$ they are not connected by an edge, that is $(u, v) \notin E$.

A set S is said to be **maximum** independent set if S is an independent set and largest in size. That is, if for every possible independent set $H \subseteq V$, $|S| \geq |H|$.

Assume that G is a tree.

a. Suggest a greedy algorithm that runs in linear time and finds a maximum independent set in G.

b. State the GCP claim corresponding to your algorithm.

c. Prove the correctness of the GCP claim from item (b) for your algorithm.

A set S is said to be **maximal** independent set if S is an independent set and for every vertex that is not in S ($u \in V, u \notin S$) the set $S \cup \{u\}$ is not an independent set.

d. Give an example of a graph $G = (V, E)$ with $V = \{a, b, c\}$, such that G has a maximal independent set, S_1, of size $|S_1| = 1$ and a maximum independent set S_2, of size $|S_2| = 2$.

Problem 2 (10 pts):

a. Given is a text of length 100 over the alphabet $\Sigma = \{a, b, c, d, e, f, g\}$. The number of appearances of every letter in the text is given in the following table.

a	b	c	d	e	f	g
30	20	14	12	10	8	6

Find an optimal code for the text.

1. Describe the binary tree that corresponds to the code

2. Complete the table with the code of each letter.

a	b	c	d	e	f	g

Problem 3 (18 pts):

Prove or give a counter example for each of the following claims.

a. If all letters have the same frequency, then Huffman algorithm produces a fixed-length code.

b. If for two letters $a, b \in \Sigma$ it holds that $p(a) \leq p(b)$ then in any optimal Huffman coding given by a tree T, $d_T(a) \leq d_T(b)$ (the depth of a in T is equal to or smaller than the depth of b in T).

c. If Huffman produces a fixed length code then for any two letters $a, b \in \Sigma$, $\frac{1}{2} \leq p(a)/p(b) \leq 2$.

Problem 4 (24 pts):

As the cold winter season begins, heavy rains start pouring on the Negev desert. Despite preparations, severe floods catch Netivei Israel (National Transport Infrastructure Company) by surprise. An urgent request goes out for a plan to position aid teams along the country's main southern artery.

Route 90, the longest road in Israel, is now a top priority. Along the portion stretching from the Dead Sea to Eilat, n flood events have already been recorded. Each flood event is associated with a segment $[s_i, t_i]$ representing the kilometer interval of the road that may have been affected. Note that a point on the road may unfortunately belong to multiple such segments.

An emergency team can be parked at any kilometer point x along the road. A team parked at x can effectively handle any flood prone segment $[s_i, t_i]$ that contains x ($s_i \leq x \leq t_i$).

Netivei Israel company seeks to deploy as few teams as possible, while still ensuring that every flood-prone segment contains at least one team.

a. Propose an algorithm that, given the set of segments $\{[s_i, t_i]\}_{i=1}^{n}$, determines where to place teams in order to minimize the total number of teams required.

b. Prove that your algorithm is optimal.

c. Analyze your algorithm's runtime complexity.

Problem 5 (24 pts):

A music festival has two stages running simultaneously: the **Rock Stage** and the **Jazz Stage**. There are $N_R>0$ seats available at the Rock Stage, and $N_J>0$ seats available at the Jazz Stage, with $N_R+N_J=n$.

n people arrive at the festival. Every person i gives an integer enjoyment score r_i if they attend Rock and j_i if they attend Jazz. Offer the festival manager an efficient greedy algorithm that divides the attendees into two groups R and J, so that $|R| = N_R$, $|J| = N_J$, group R will be directed to the Rock stage, group J will be directed to the Jazz stage, and their total satisfaction, given by $\sum_{i\in R} r_i + \sum_{i\in J} j_i$ will be maximal.

1. Describe the algorithm. State and justify its time complexity.
2. State and prove the corresponding greedy choice property.